

Name: _____

Date: _____

RATIO, (RATIO AND PROPORTION PROBLEMS)

LO: I can solve problems involving ratio and proportion in context.

Ratio and proportion problems

A **ratio** compares two or more quantities. **Proportion problems** ask us to use a known ratio to find unknown amounts.

Method: Find the value of one part first, then scale up or down to answer the question.

Quick examples

●	Purple paint: red to blue = 3 : 5	for every 3 parts red, use 5 parts blue
●	If 4 books cost £12, then 7 books cost £21	find the cost of one book first
●	Share £40 in ratio 3 : 5 gives £15 and £25	8 parts total, one part = £5
●	3 : 5 means 3 out of 5	it means 3 out of 8 total parts

Key idea

Find the total number of parts, then divide the quantity by the total to find the value of one part.

$$\text{one part} = \text{total} \div$$

$$\text{sum of ratio parts}$$

Then multiply one part by each side of the ratio

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - sharing money

Share £60 in the ratio 2 : 3

$$\text{Total parts} = 2 + 3 = 5$$

$$\text{One part} = £60 \div 5 = £12$$

$$2 \text{ parts} = £24, 3 \text{ parts} = £36$$

£24 and £36

Always check the shares add to the original total.

Example 2 - scaling a recipe

A recipe for 4 people uses 300 g flour.
How much for 6 people?

$$\text{Find amount per person: } 300 \div 4 = 75 \text{ g}$$

$$\text{Multiply for 6: } 75 \times 6 = 450 \text{ g}$$

Answer: 450 g of flour

450 g

Find the unitary value (for one), then scale up.

Example 3 - finding a missing quantity

Red and blue paint are mixed in ratio 2 : 5. If 10 litres of red are used, how much blue?

$$2 \text{ parts} = 10 \text{ litres, so } 1 \text{ part} = 5 \text{ litres}$$

$$\text{Blue} = 5 \text{ parts} = 5 \times 5 = 25 \text{ litres}$$

$$\text{Check: } 10 : 25 = 2 : 5$$

25 litres

Use the known quantity to find one part, then find the unknown.

Example 4 - three-part ratio

Share 72 sweets in the ratio 1 : 2 : 3

$$\text{Total parts} = 1 + 2 + 3 = 6$$

$$\text{One part} = 72 \div 6 = 12$$

$$\text{Shares: } 12, 24, \text{ and } 36$$

12, 24, and 36

Three-part ratios work the same way as two-part ratios.

YOUR TURN – PRACTICE (deliberate practice)

1. Sharing in a ratio

Share each amount in the given ratio.

- a) £30 in ratio 1 : 2
- b) £56 in ratio 3 : 4
- c) 80 kg in ratio 2 : 3
- d) £100 in ratio 3 : 7

2. Scaling recipes

Use proportion to find the missing amount.

- a) 5 cakes need 200 g sugar. How much for 8 cakes?
- b) 3 shirts cost £45. How much for 7 shirts?
- c) 4 tins paint cover 20 m². How many for 35 m²?
- d) 6 workers finish in 12 days. How long for 9 workers?

3. Finding missing quantities

Use the ratio to find the unknown.

- a) $a : b = 3 : 5$. If $a = 21$, find b .
- b) $x : y = 4 : 7$. If $y = 42$, find x .
- c) Cement : sand = 1 : 4. If sand = 60 kg, find cement.
- d) Boys : Girls = 5 : 3. If 20 boys, find girls.

4. Three-part ratios

Share in the given three-part ratio.

- a) £60 in ratio 1 : 2 : 3
- b) 120 cm in ratio 2 : 3 : 5
- c) £150 in ratio 1 : 2 : 2
- d) 90 kg in ratio 3 : 4 : 2

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To share £40 in ratio 3 : 5, find total parts = 8, one part = £5 ☐

Correct answer: _____

Reason: _____

b) The ratio 3 : 5 means there are 5 items in total ☐

Correct answer: _____

Reason: _____

c) Doubling a recipe means doubling all ingredients ☐

Correct answer: _____

Reason: _____

d) In ratio 2 : 3, the first quantity is always larger ☐

Correct answer: _____

Reason: _____

e) The unitary method finds the value of one part first ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A fruit drink is made by mixing orange juice and lemonade in the ratio **2 : 5**. Sam makes **2.1 litres** of the drink. How much of each ingredient does he use? If orange juice costs **£1.20 per litre** and lemonade costs **80p per litre**, what is the total cost?

Expression: _____

Simplified: _____

b) Three friends share a prize in the ratio **3 : 4 : 5**. The largest share is **£15 more** than the smallest share. How much is the total prize? If they then donate $\frac{1}{4}$ of each share to charity, how much does each donate?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

